Aga Khan University Examination Board

Notes from E-Marking Centre on SSC-II Physics Examination May 2018

Introduction:

This document has been produced for the teachers and candidates of Secondary School Certificate (SSC) Part II Physics. It contains comments on candidates' responses to the 2018 SSC Examination, indicating the quality of the responses and highlighting their relative strengths and weaknesses.

E-Marking Notes:

This includes overall comments on candidates' performance on every question and *some* specific examples of candidates' responses which support the mentioned comments. Please note that the descriptive comments represent an overall perception of the better and weaker responses as gathered from the e-marking session. However, the candidates' responses shared in this document represent some specific example(s) of the mentioned comments.

Teachers and candidates should be aware that examiners may ask questions that address the Student Learning Outcomes (SLOs) in a manner that requires candidates to respond by integrating knowledge, understanding and application skills they have developed during the course of study. Candidates are advised to read and comprehend each question carefully before writing the response to fulfil the demand of the question.

Candidates need to be aware that the marks allocated to the questions are related to the answer space provided on the examination paper as a guide to the length of the required response. A longer response will not in itself lead to higher marks. Candidates need to be familiar with the command words in the SLOs which contain terms commonly used in examination questions. However, candidates should also be aware that not all questions will start with or contain one of the command words. Words such as 'how', 'why' or 'what' may also be used.

General Observations:

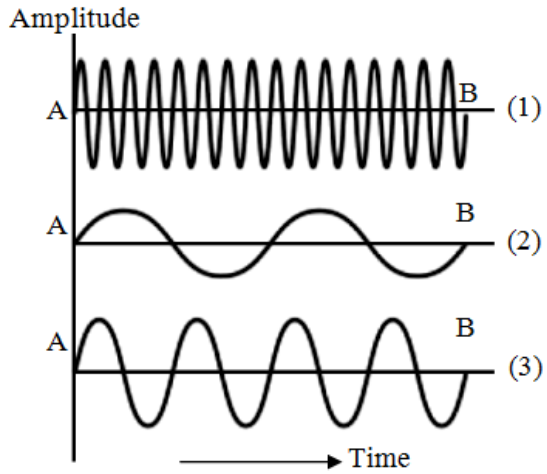
This year, candidates performed well on questions related to the refraction of light, total internal reflection, turning effect on a current carrying coil in a magnetic field, flow of information, waves, their nature and types, noise pollution, image location by lens equation, capacitors and capacitance. Whereas low-scoring candidates did not perform well in questions based on making and completing ray diagrams of the transparent materials, word problems based on the characteristics of waves, safety measures, electric field and its intensity, natural radioactivity, resistance (series and parallel combinations) and also had problem in correctly articulating their answers on the basic definition, reasoning and discussion type questions.

Detailed Comments:

Constructed Response Questions (CRQs)

Question 1:

The given diagram shows three waves covering the same distance from points (A) to (B) in 2 seconds.

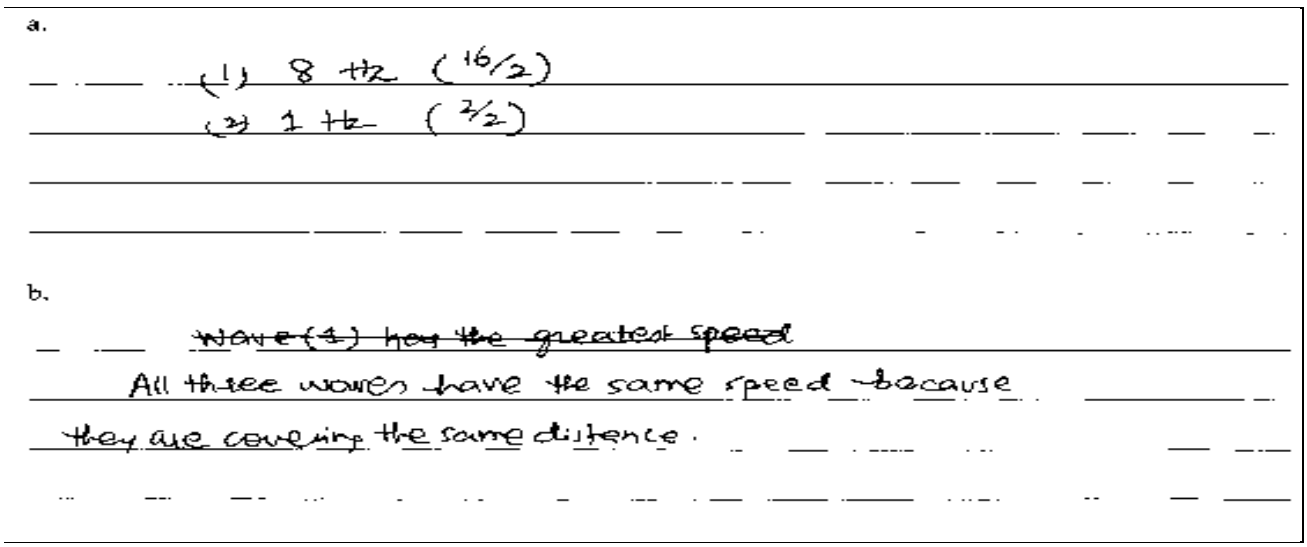


a. Write the frequency of any TWO waves.

b. Which of the labelled waves, (1), (2) or (3), has the greatest speed and why?

Better responses mentioned the correct frequency of both waves mentioned in the question, as wave (1) has 8, wave (2) has 1 and wave (3) has 2 frequencies. In the second part of the question, high scoring candidates correctly mentioned that all three waves have the same speed because all waves cover same distance in 2 seconds.

Example:


a.

(1) 8 Hz (16/2)

(2) 1 Hz (2/2)

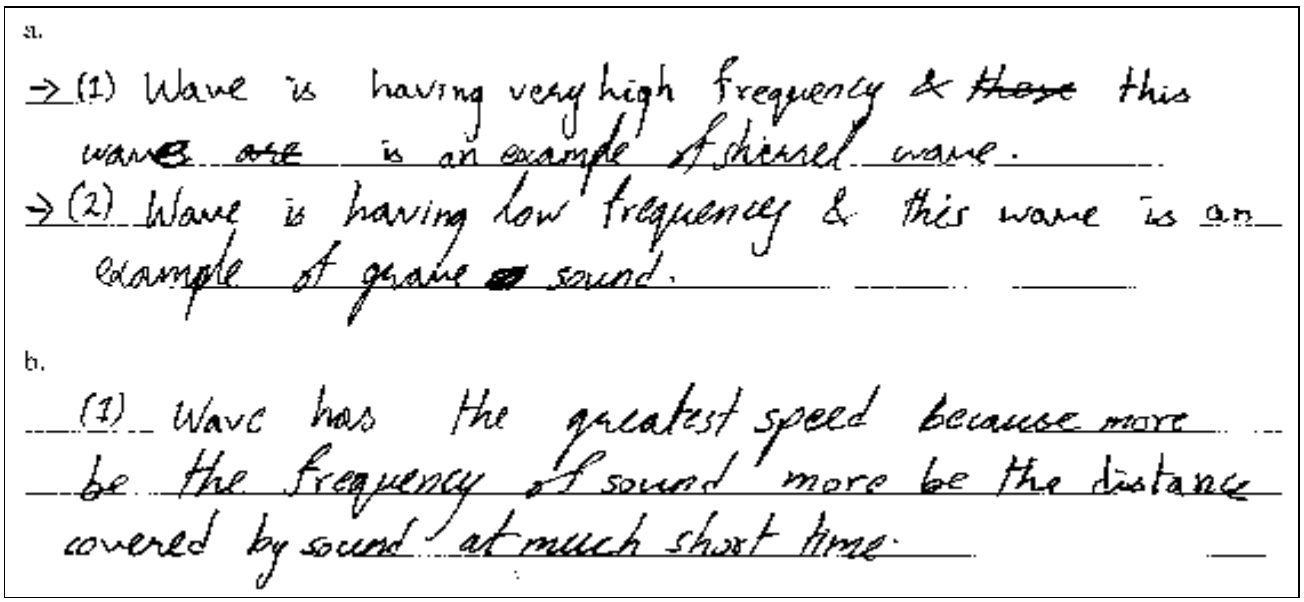
b.

~~Wave (1) has the greatest speed~~

All three waves have the same speed because they are covering the same distence.

Weaker responses showed that candidates lacked understanding of calculating the correct frequency of both waves, some low scoring candidates wrote the frequencies as low and high. In the second part of the question, candidates wrote sound and light waves which were irrelevant of the demand of the question.

Example:


a.

→ (1) Wave is having very high frequency & ~~those~~ this wave ~~are~~ is an example of shrill wave.

→ (2) Wave is having low frequency & this wave is an example of grave sound.

b.

(1) Wave has the greatest speed because more be the frequency of sound more be the distance covered by sound at much short time.

Question 2:

Identify the following as 'True' OR 'False' and give a reason to justify any ONE of your answers.

a. Light travels slower through glass than empty space. _____

b. When light travels from glass to air, its path bends towards the normal. _____

Better responses correctly identified part (a) as true and part (b) as false and correctly justified their identification. For example, in part (a), they wrote that light refracts when it encounters different mediums and the refractive index shows the relationship between the speed of light in empty space and the speed when it travels through a material that is why light travels slower through glass than empty space. In part (b), high scoring candidates wrote that refraction is the bending of the path of a light wave as it passes from one material into another material. The refraction occurs at the boundary and is caused by a change in the speed of the light. Therefore, the path of a light wave bends depends on whether the light wave is travelling from a denser/ slow medium to a less dense/ fast medium.

Example:

a. Light travels slower through glass than empty space. True

b. When light travels from glass to air, its path bends towards the normal. false

Justification:

·Yes ✓ ligh travel slow in denser ~~one~~ medium and fast in empty in empty space.

· When light travel from denser to rare medium it donot bends toward normal

Weaker responses revealed that candidates were unable to identify both parts of the question. But few of the low scoring candidates had successfully justified their identification.

Example:

a. Light travels slower through glass than empty space. false.

b. When light travels from glass to air, its path bends towards the normal. True.

Justification:

(b) because when any thing pass through denser to rare medium it can pass towards the normal.

Question 3:

Mention TWO conditions for the occurrence of total internal reflection.

Better responses correctly mentioned two conditions for the occurrence of total internal reflection, like the one shown in the below example of better response image.

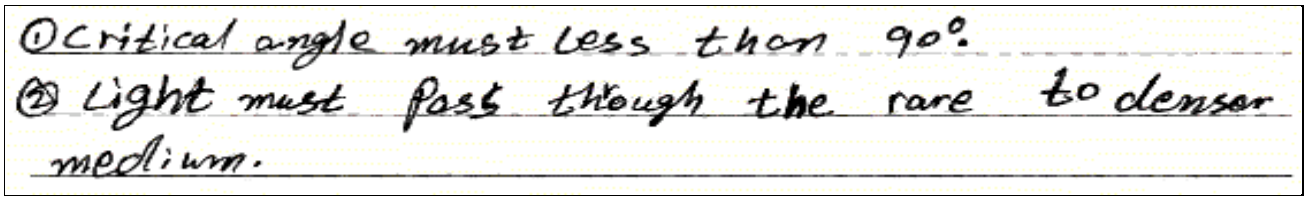
Example:

① The light ray must travel from denser to rare medium

② The angle of incidence must be greater the critical angle.

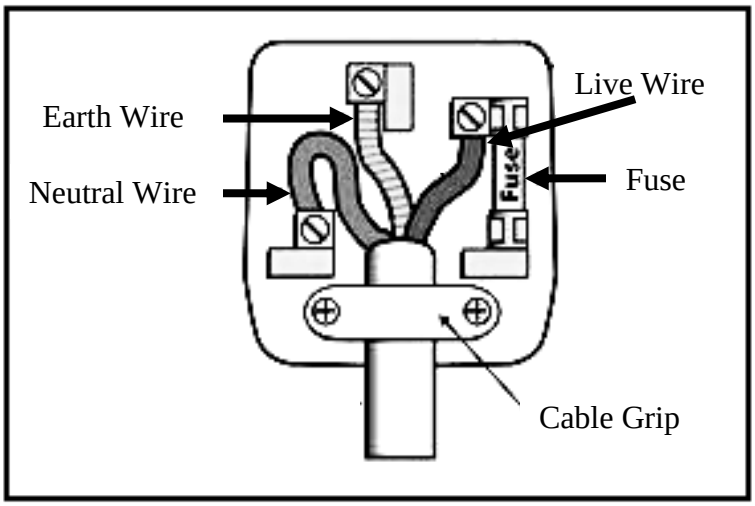
Weaker responses showed candidates' inability to write the conditions for the occurrence of total internal reflection. Some of the low scoring candidates wrote about the concept of critical angle, angle of incidence and angle of refraction. Few candidates incorrectly wrote that the light passes through rare to denser medium.

Example:



Question 4:

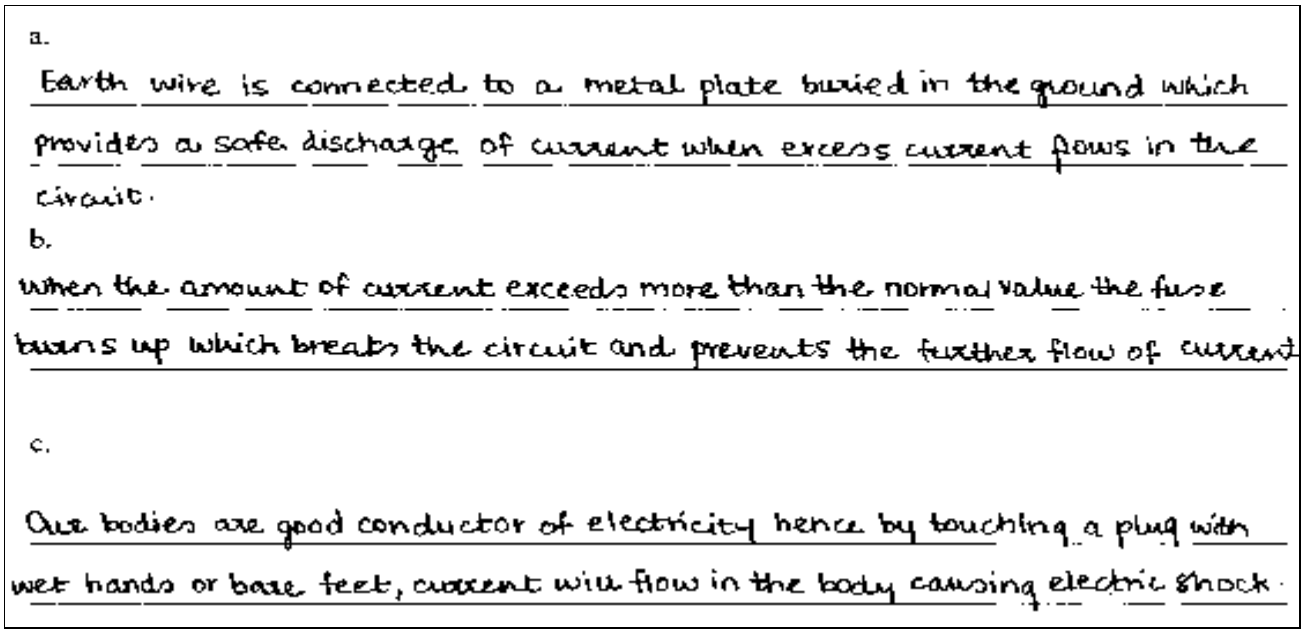
The given figure shows a power plug.



a. State the function of the earth wire shown in the given figure.
b. Describe the working of a fuse in the power plug.
c. What would happen when one touches a power plug with wet hands or when one is barefoot?

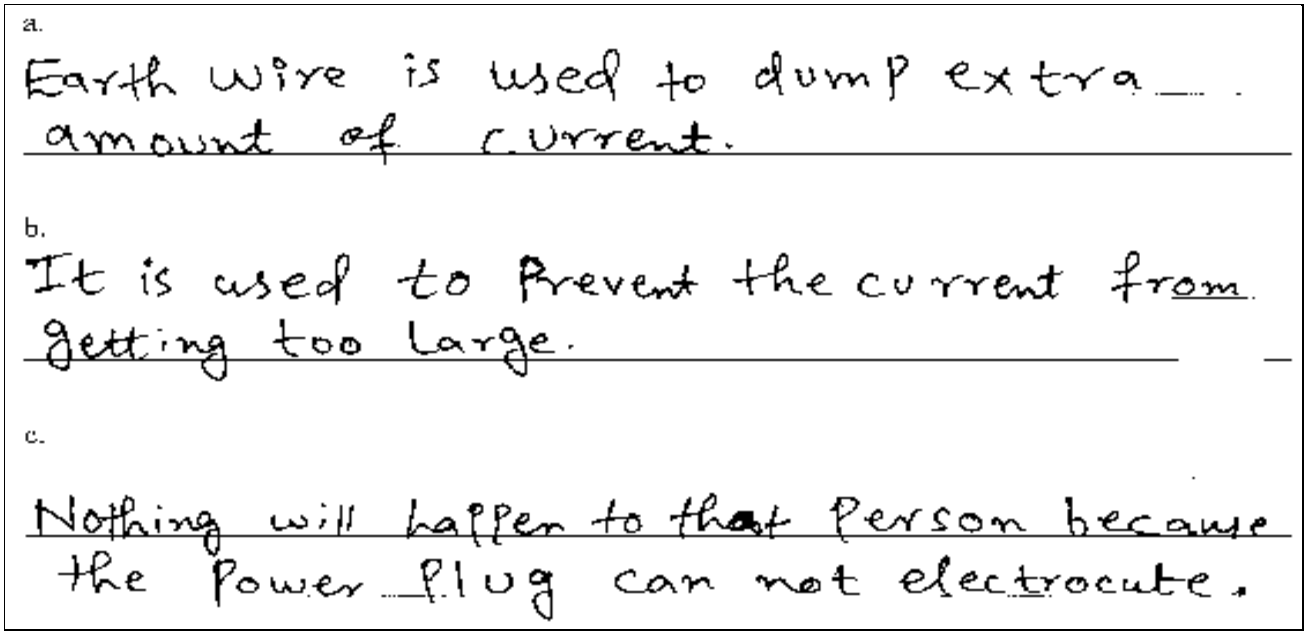
Better responses correctly stated the function of the earth wire by writing that it protects the user from electric shock and drains the static charges that might be present on the body of the appliance to the earth.
In the second part of the question, high scoring candidates correctly described about the working of fuse in the power plug by writing that a safety device that is connected in series with the live wire in the circuit that can protect equipments when excess current flows and if a large, unsafe current passes through the electric circuit, the fuse melts and breaks the circuit before the wires become very hot and cause fire.
In the third part of the question, they wrote that circuit gets completed between the conductors in the switch and wet hands through the waters. When someone touched the switch only with wet hands, there are little chances that the water will not be enough to seep into the switch and get the circuit complete to be electrocuted but barefooted on the ground can give the current an escape way to complete electric circuit, and a person could be electrocuted.

Example:


a.
Earth wire is connected to a metal plate buried in the ground which provides a safe discharge of current when excess current flows in the circuit.
b.
when the amount of current exceeds more than the normal value the fuse burns up which breaks the circuit and prevents the further flow of current
c.
Our bodies are good conductor of electricity hence by touching a plug with wet hands or base feet, current will flow in the body causing electric shock.

Weaker responses correctly stated and described the function of the earth wire and the working of a fuse in the power plug respectively. Few of the low scoring candidates were unable to reason out that what will happen, when one touches a power plug with wet hands or when one is barefoot, they wrote that nothing will happen and a person could not be electrocuted.

Example:



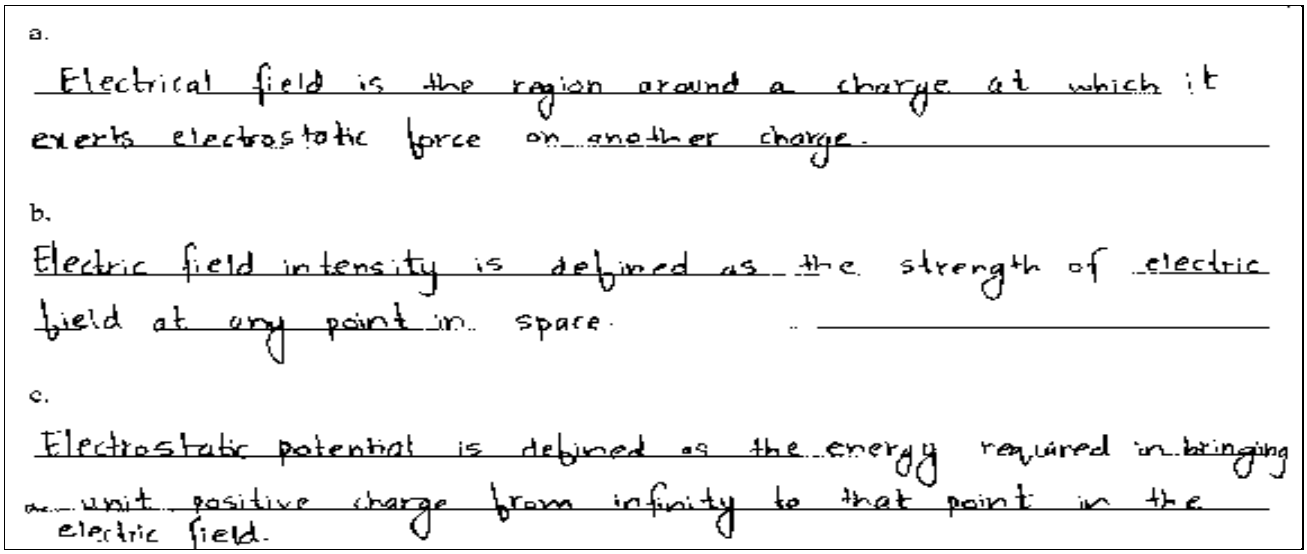
Question 5:

Define the following:

a. Electric field
b. Electric field intensity
c. Electrostatic potential

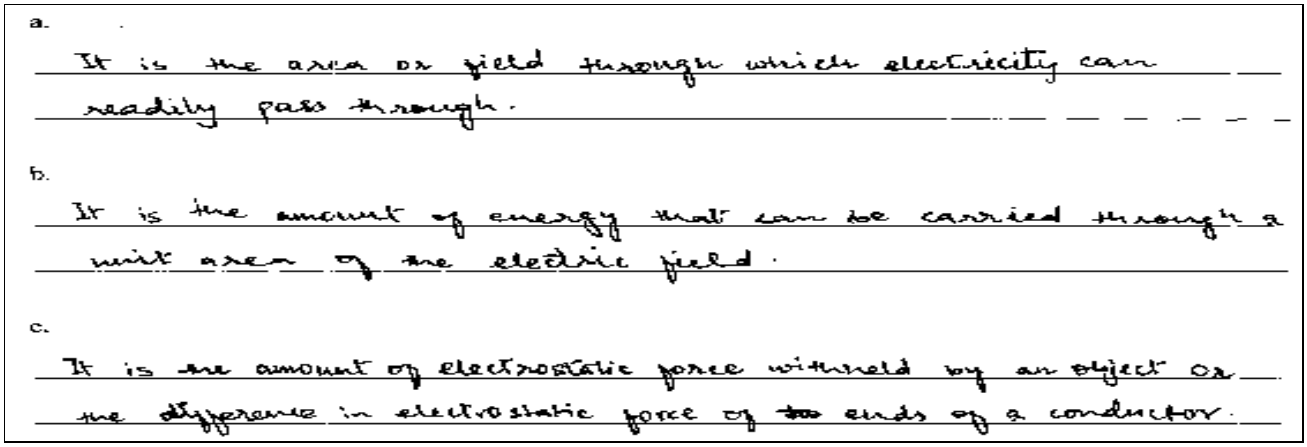
Better responses correctly defined the electric field, electric field intensity and electrostatic potential as the one shown in the below example of better response image.

Example:



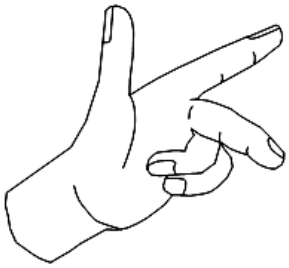
Weaker responses reflected that candidates lacked knowledge about the basic concepts of electric field, electric field intensity and electrostatic potential. They wrote about electricity, conductors and insulators which are irrelevant to the demand of the question.

Example:


a.
It is the area or field through which electricity can readily pass through.
b.
It is the amount of energy that can be carried through a unit area of the electric field.
c.
It is the amount of electrostatic force withheld by an object or the difference in electrostatic force of the ends of a conductor.

Question 6:

The given figure shows the direction of the magnetic field, the current and the force on a conductor according to Fleming's left hand rule.

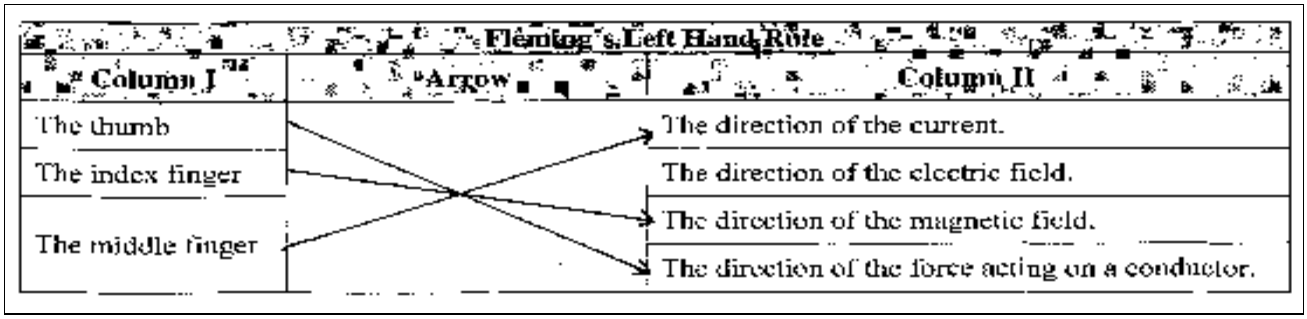


Match the items in column **I** with the items in column **II** using arrows (⟶).

Fleming's Left Hand Rule		
Column I	**Arrow**	**Column II**
The thumb		The direction of the current.
The index finger		The direction of the electric field.
The middle finger		The direction of the magnetic field.
		The direction of the force acting on a conductor.

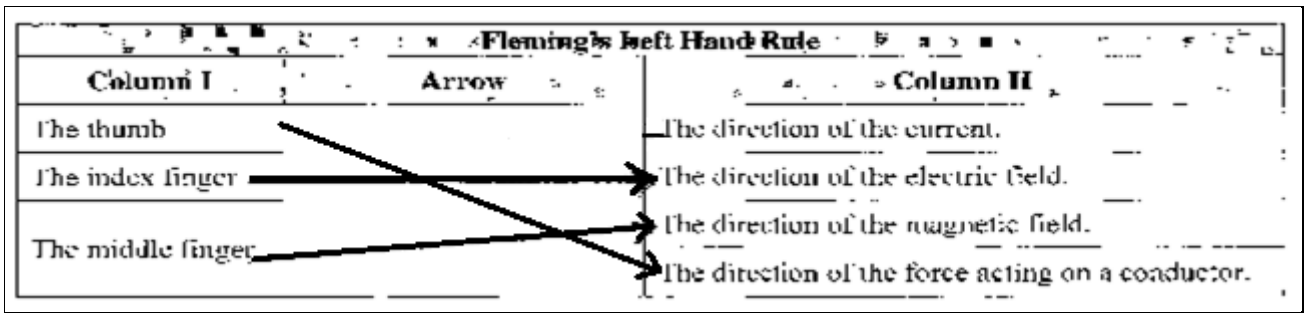
Better responses correctly matched the items in column I with the items in column II using arrows (⟶) as the one shown in the below example of better response image.

Example:



Weaker responses were unable to match correctly all three items of column I with the items in column II that shows their lack of conceptual understanding and in-depth knowledge. Some of the low scoring candidates matched only one item correctly.

Example:

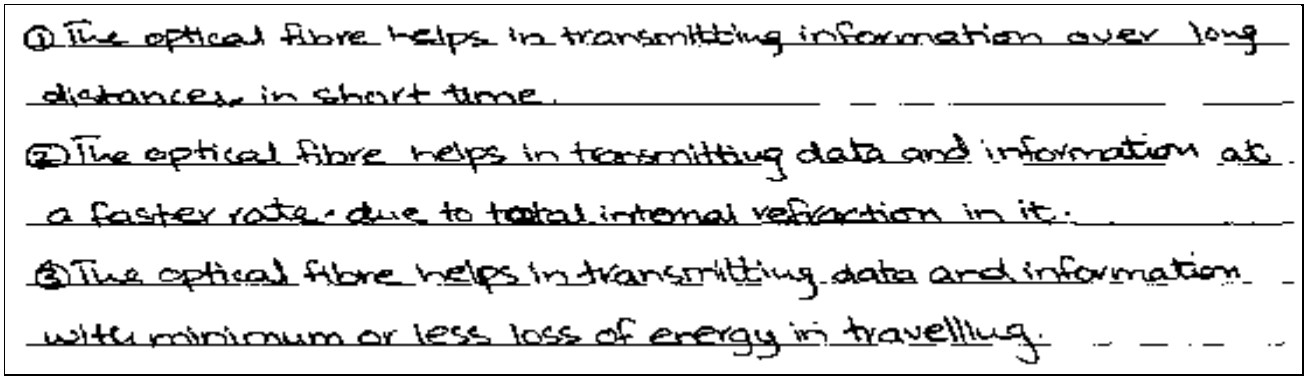


Question 7:

Mention any THREE benefits of transmitting information and data through optical fibre.

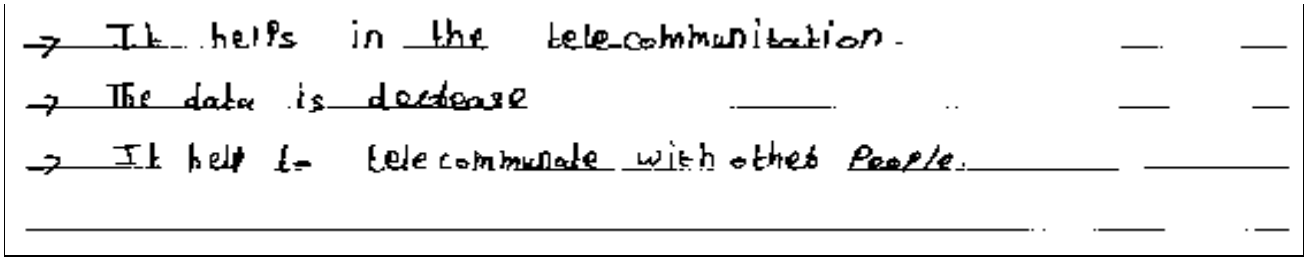
Better responses correctly mentioned three benefits of transmitting information and data through optical fibre. For example, many signals can be sent and received at a time, while transmitting light signals, there is no loss of energy due to total internal reflection, light signals transmitted by optical fibre travel very fast and optical fibre is very light in weight, therefore it is easy to install.

Example:



Weaker responses failed to write the correct benefits of transmitting information and data through optical fibre. Few low scoring candidates wrote about the telecommunication, database, internet, Wi-Fi and other mobile application softwares.

Example:



Question 8:

Reaction # 1:

$^{218}Po_{84} \; - \; ^{4}He_{2} \; = \; ^{214}Pb_{82}$

Reaction # 2:

$^{214}Pb_{82} \; - \; ^{0}e_{-1} \; = \; ^{214}Bi_{83}$

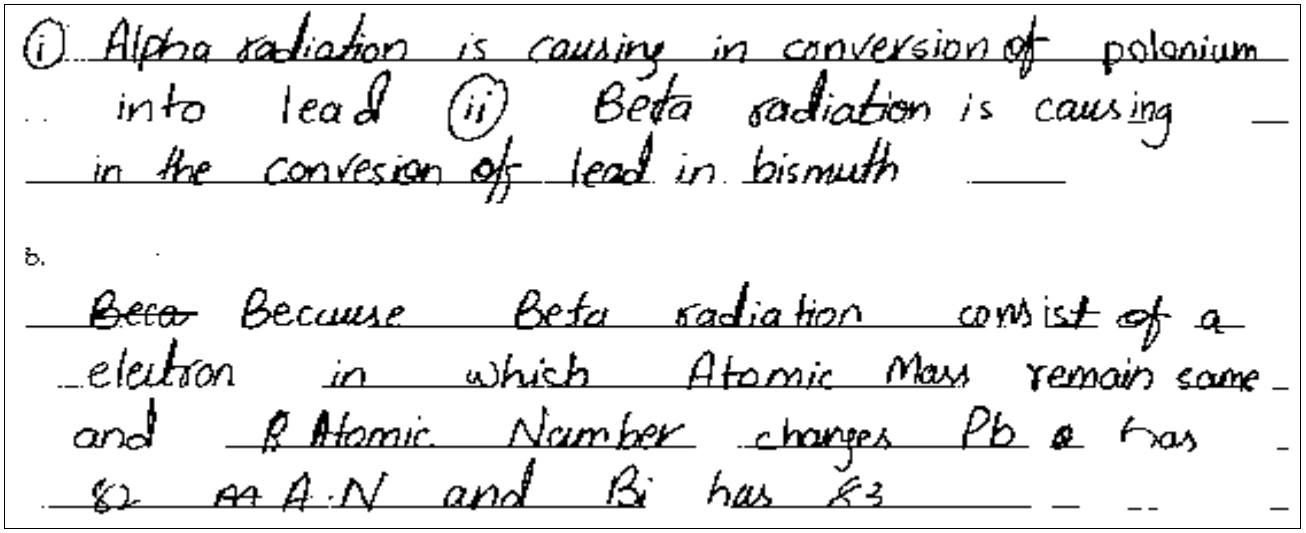
a. Which type of nuclear radiation causes the following changes?

   i. Conversion of polonium into lead in reaction # 1
   ii. Conversion of lead into bismuth in reaction # 2

b. Give a reason to support your answer for (ii) of part (a).

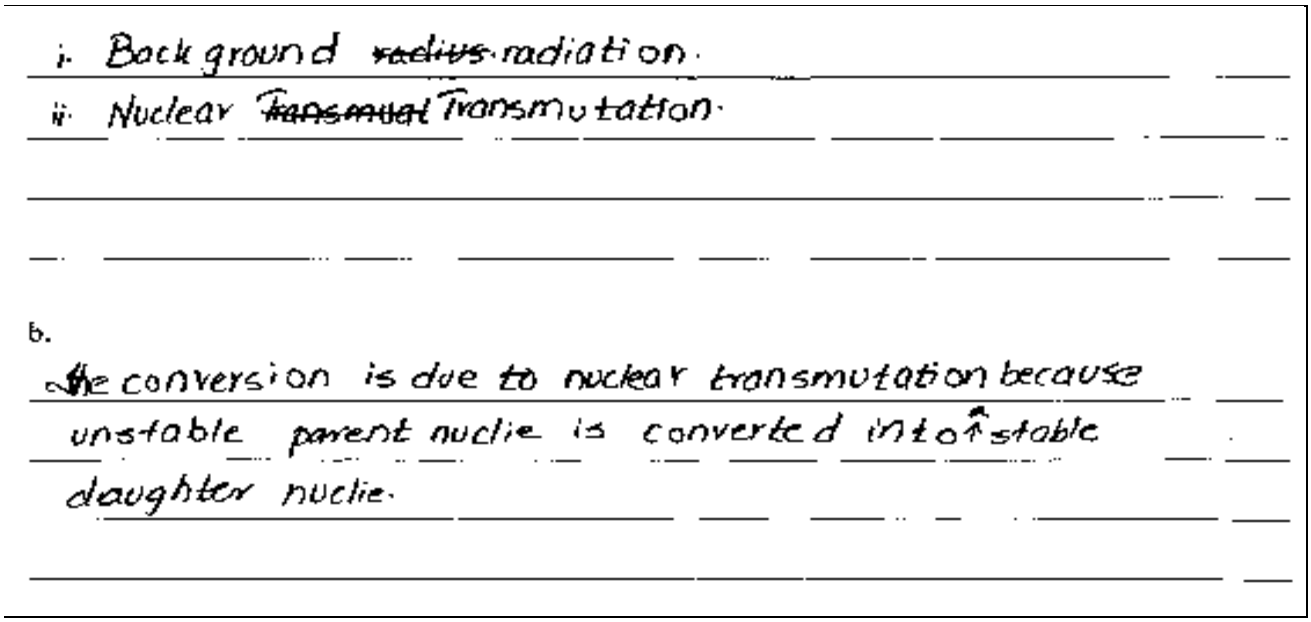
Better responses correctly wrote the type of nuclear radiation that causes the conversion of polonium into lead in reaction # 1 and conversion of lead into bismuth in reaction # 2 as the emission of an alpha particle and the emission of a beta particle respectively. In the second part of the question, high scoring candidates reason out in support of their answer for (ii) of part (a) as a neutron in the nucleus has changed into a proton by emitting an electron and this leaves one more proton and one fewer neutron in the nucleus and therefore nucleon number does not change, but the proton number or charge number increase by 1 unit.

Example:



Weaker responses incorrectly wrote their answers and mentioned background radiation, nuclear transmutation, nuclear decay etc. in the first part of the question. In the second part, low scoring candidates wrote about nuclear transmutation, parent and daughter elements, stable and unstable nucleus which are completely irrelevant to the demand of the question.

Example:



Extended Response Questions (ERQs)

These questions offered a choice between part **a** and **b**

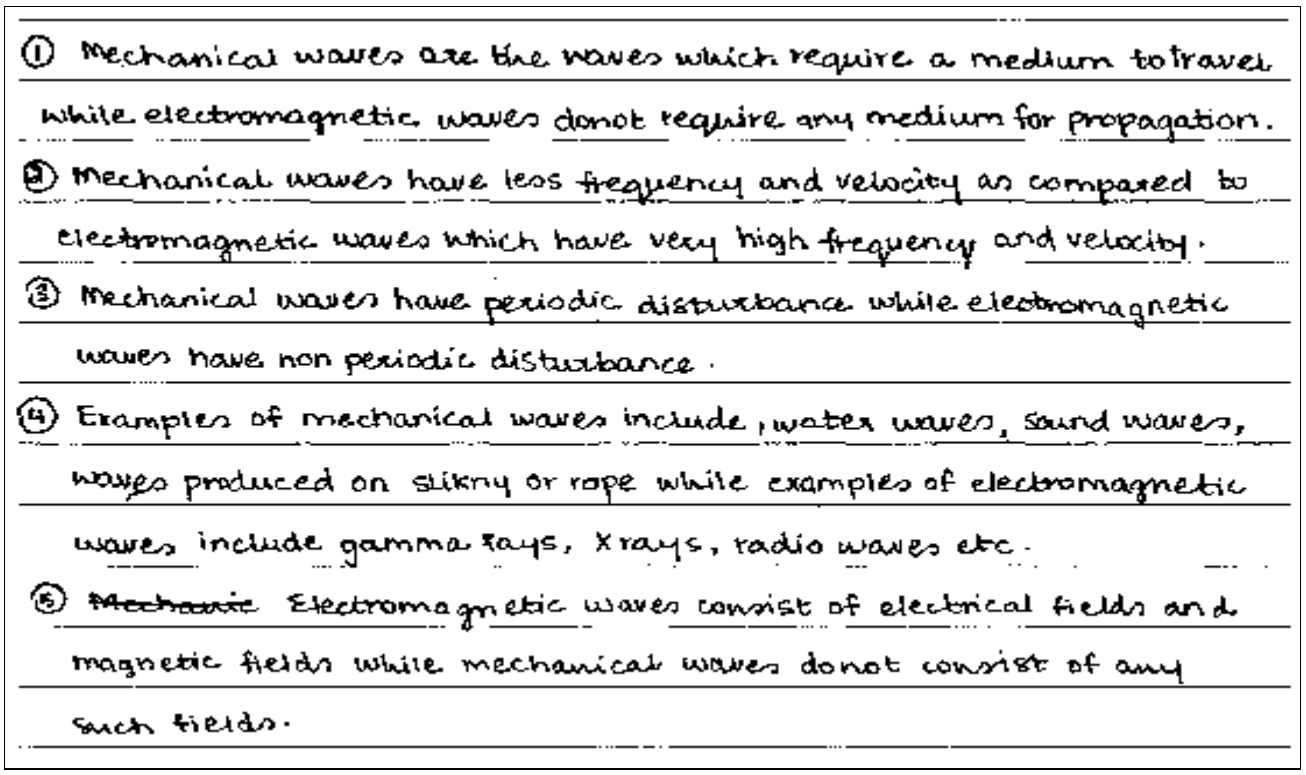
(**Note**: Majority of the candidates attempted Question **9a** and they scored well in this part.)

Question 9a:

Describe any FIVE differences between mechanical and electromagnetic waves.

Better responses correctly described five differences between mechanical and electromagnetic waves. High scoring candidates wrote the differences between the waves by mentioning in their answers the following points. For example, material medium for the propagation, vibrations of the particles of the medium, concept of speed, particle like behaviour, affected by an electrical and magnetic fields, periodic disturbance, energy of the electromagnetic waves, display particle or wave like behaviour and examples as X-rays, radio waves, water waves, sound waves etc.,

Example:



① Mechanical waves are the waves which require a medium to travel while electromagnetic waves donot require any medium for propagation.

② Mechanical waves have less frequency and velocity as compared to electromagnetic waves which have very high frequency and velocity.

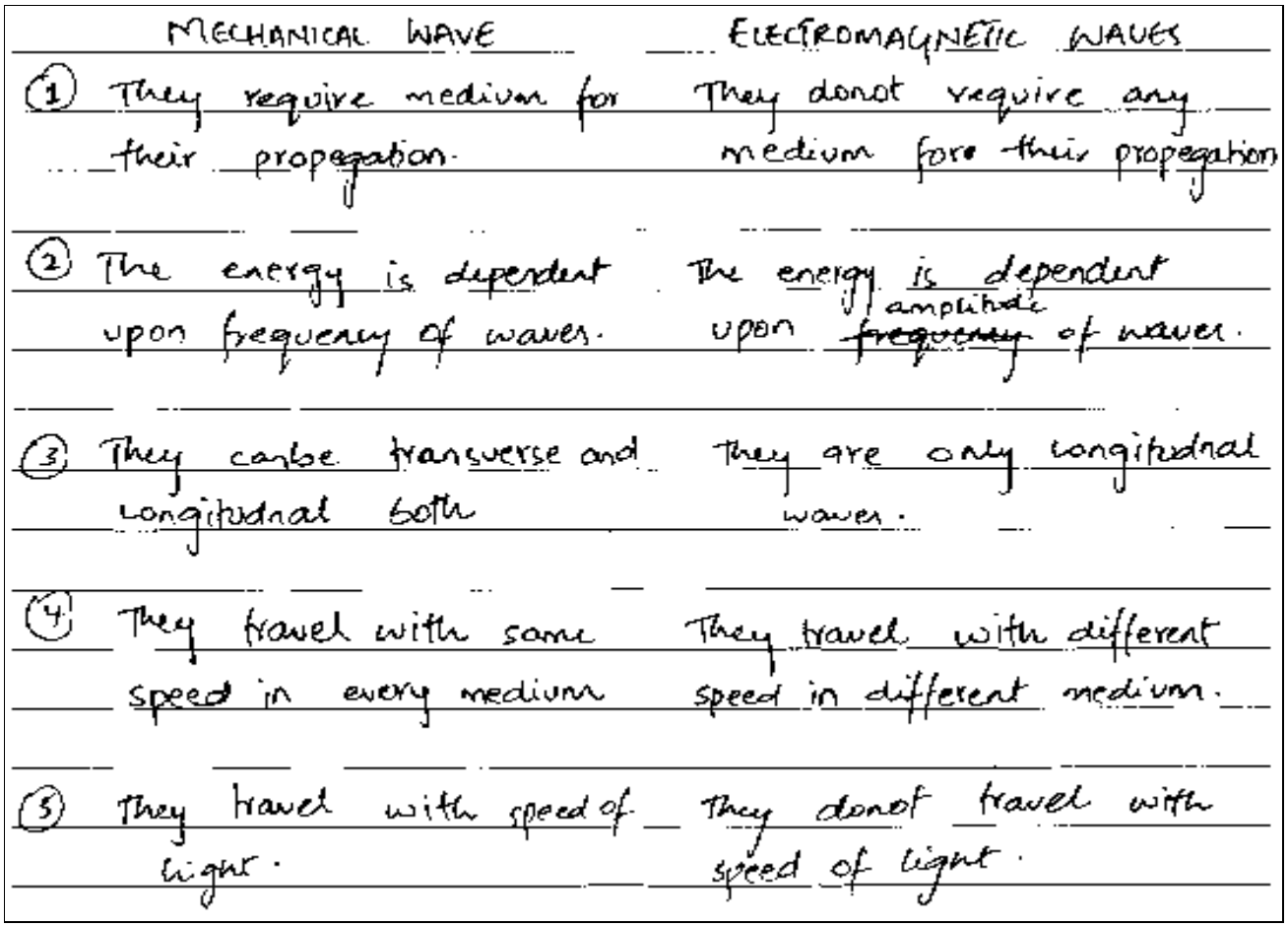
③ Mechanical waves have periodic disturbance while electromagnetic waves have non periodic disturbance.

④ Examples of mechanical waves include, water waves, sound waves, waves produced on slinky or rope while examples of electromagnetic waves include gamma rays, X rays, radio waves etc.

⑤ ~~Mechanic~~ Electromagnetic waves consist of electrical fields and magnetic fields while mechanical waves donot consist of any such fields.

Weaker responses revealed that some candidates wrote only two or three differences. Some of the low scoring candidates correctly differentiate between mechanical and electromagnetic waves in terms of medium for the propagation, energy dependent, longitudinal and transverse waves and in terms of concept of speed of the waves.

Example:



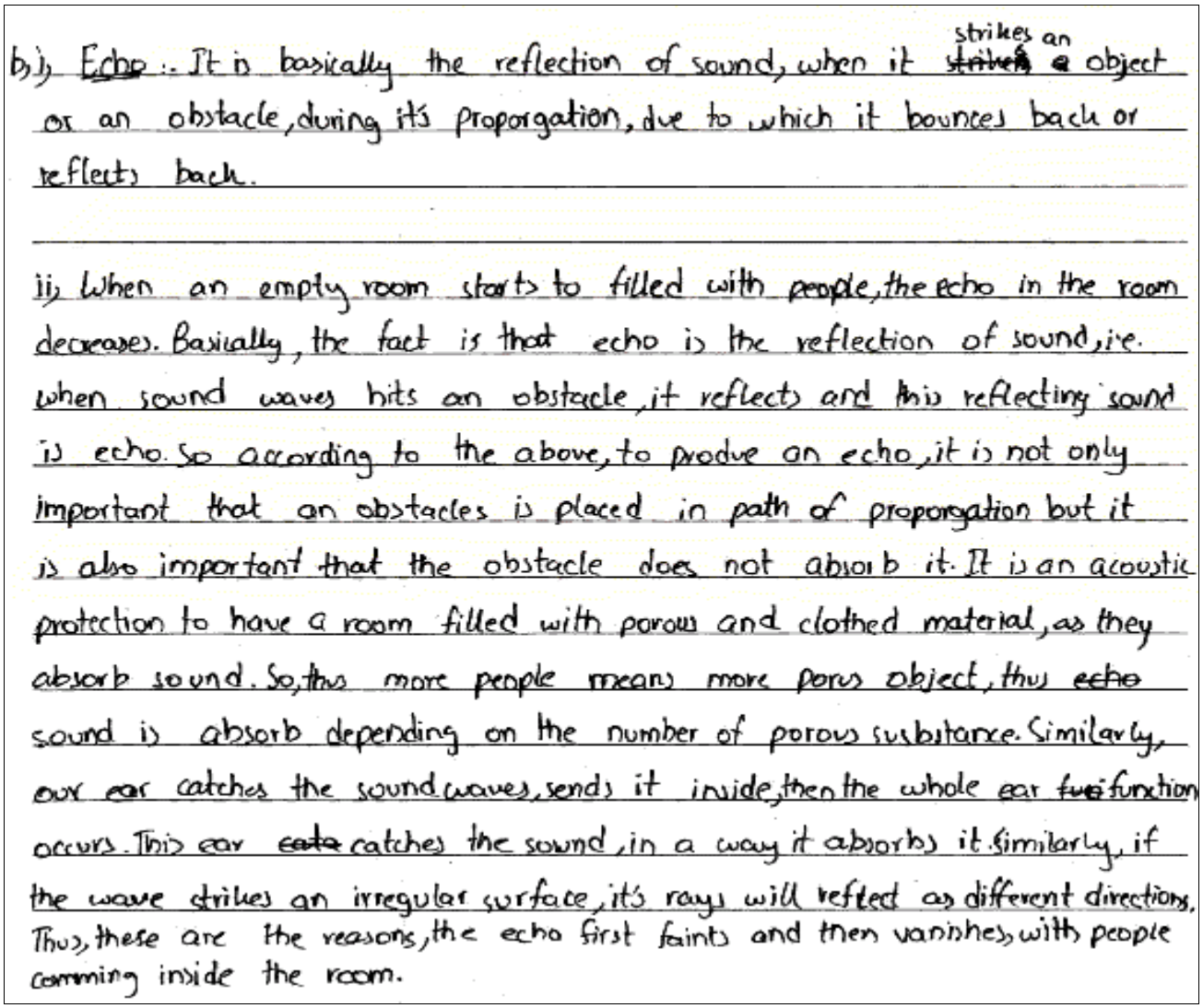
MECHANICAL WAVE	ELECTROMAGNETIC WAVES
(1) They require medium for their propagation.	They donot require any medium fore their propegation
(2) The energy is dependent upon frequency of waves.	The energy is dependent upon ~~frequency~~ amplitude of waves.
(3) They canbe transverse and longitudnal both	They are only longitudnal waves.
(4) They travel with same speed in every medium	They travel with different speed in different medium.
(5) They travel with speed of light.	They donot travel with speed of light.

Question 9b:

i. Define the term 'echo'.
ii. 'When an empty room starts to get filled with people, the echo in the room decreases.' Elaborate the given statement.

In Better responses, majority of the candidates correctly defined 'echo' as when sound incidents on the surface of a medium it bounces back into the first medium and elaborated that when an empty room starts to get filled with people, the echo in the room decreases. They wrote that when the hall is empty, there are no obstacles in between the reflection of sound other than the walls, when the hall is full of audience the sound produced undergoes multiple reflections from the people and so it overlaps with the sound produced and therefore, the listener will not be able to distinguish between the original sound and the echo.

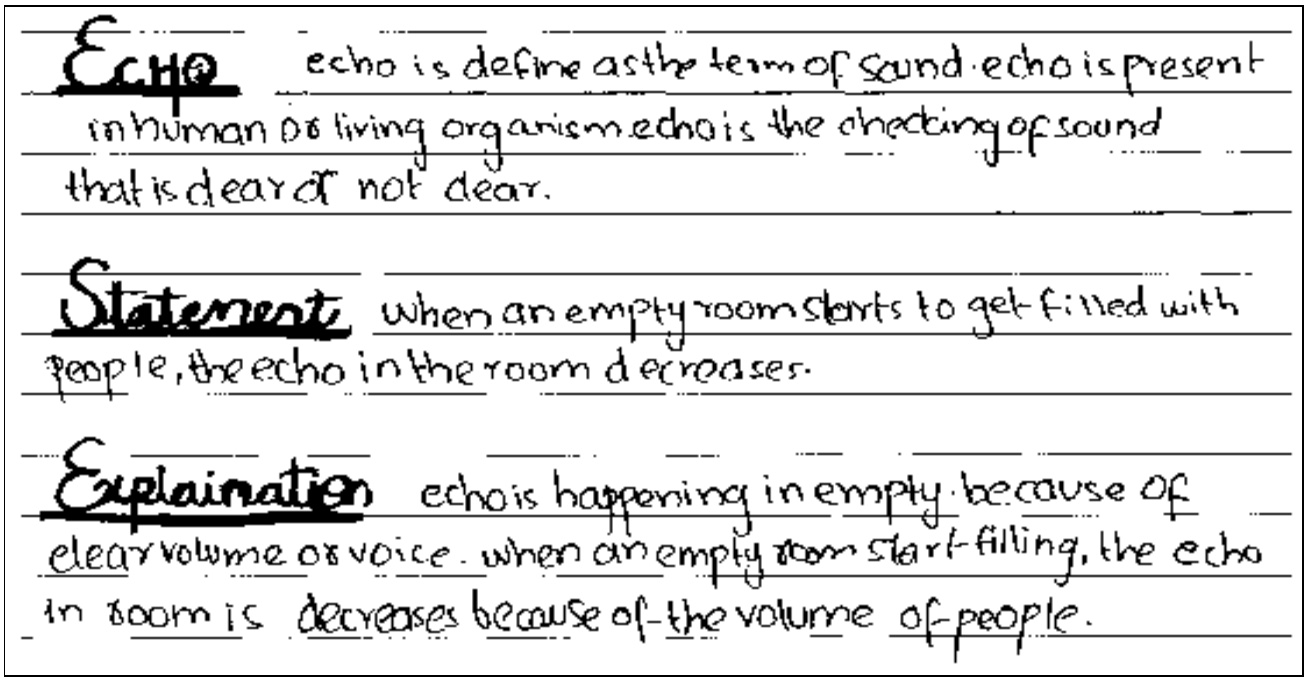
Example:


b)i) Echo :- It is basically the reflection of sound, when it strikes an object or an obstacle, during its proporgation, due to which it bounces back or reflects back.

ii) When an empty room starts to filled with people, the echo in the room decreases. Basically, the fact is that echo is the reflection of sound, i.e. when sound waves hits an obstacle, it reflects and this reflecting sound is echo. So according to the above, to produe an echo, it is not only important that an obstacles is placed in path of proporgation but it is also important that the obstacle does not absorb it. It is an acoustic protection to have a room filled with porous and clothed material, as they absorb sound. So, thus more people means more porus object, thus echo sound is absorb depending on the number of porous susbstance. Similarly, our ear catches the sound waves, sends it inside, then the whole ear ~~fuc~~ function occurs. This ear ~~eata~~ catches the sound, in a way it absorbs it. Similarly, if the wave strikes an irregular surface, it's rays will refled as different directions. Thus, these are the reasons, the echo first faints and then vanishes, with people comming inside the room.

Weaker responses reflected that candidates lacked understanding on 'echo'. They wrote about the noise and pleasant sounds for human and other living organisms. In the second part of the question, low scoring candidates were also unable to elaborate the statement about intensity of echo in an empty and filled with people room. They wrote about volume, mass and density concepts in their answers which are completely irrelevant to the demand of the question.

Example:

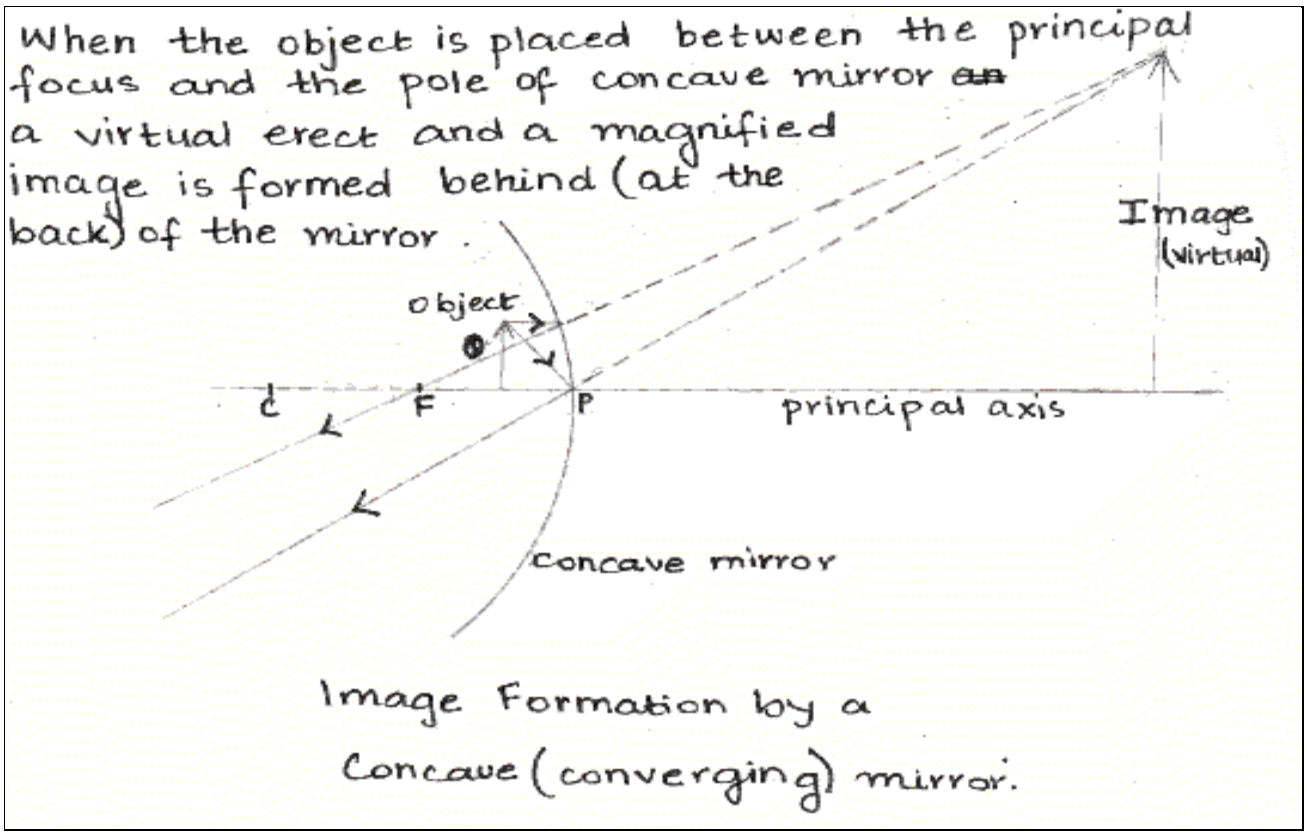


(**Note**: Majority of the candidates attempted Question **10b** and they scored well in this part.)

Question 10a:
Draw a well labelled ray diagram to illustrate the formation of an enlarged and virtual image through a converging mirror.

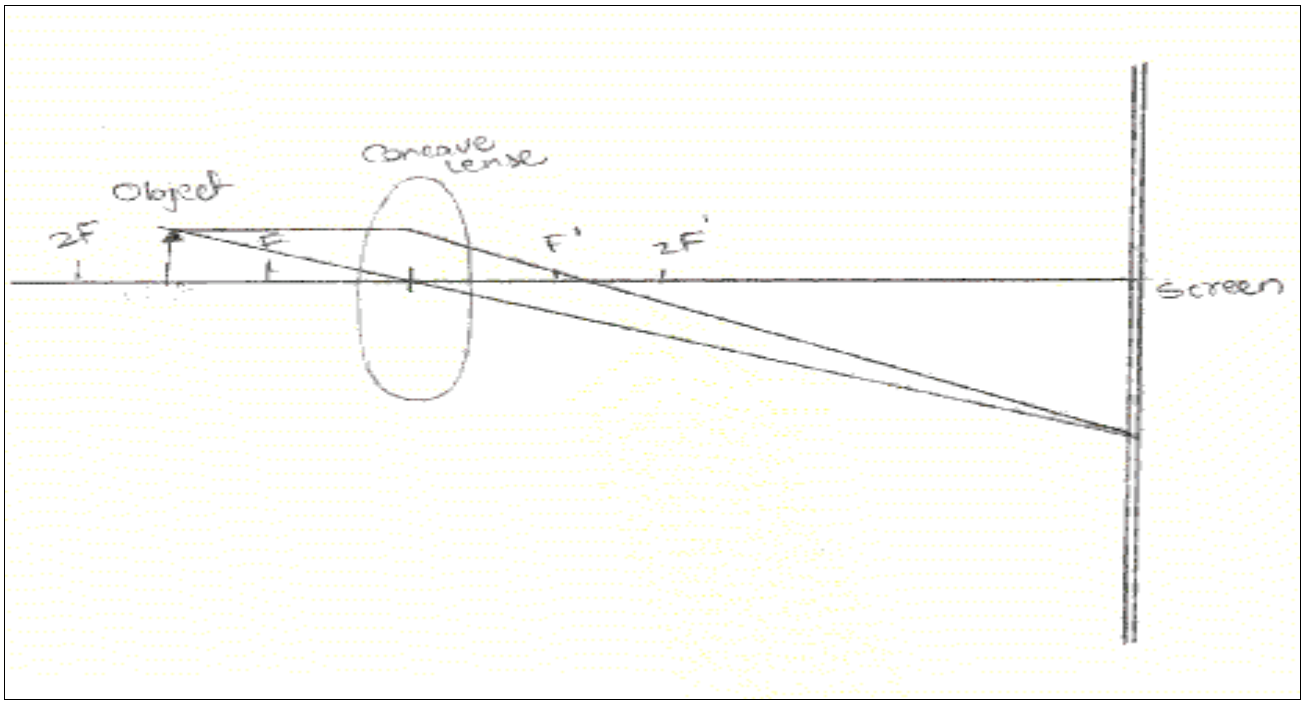
Better responses correctly draw a well labelled ray diagram that illustrated the formation of an enlarged and virtual image through a converging mirror as the one shown in the below example of better response image.

Example:



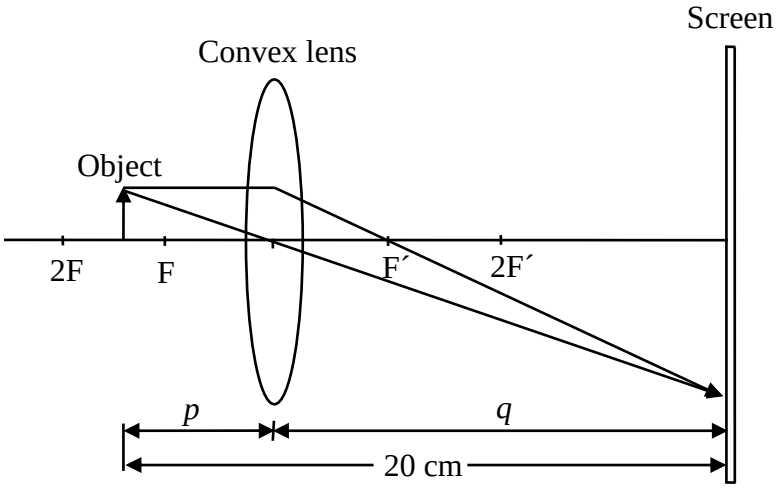
Weaker responses failed to draw a well labelled ray diagram, low scoring candidates did not know the basic difference between a mirror and a lens and were unable to place the object at the right position in front of concave mirror and ultimately failed to draw the image as demanded in the question.

Example:



Question 10b:

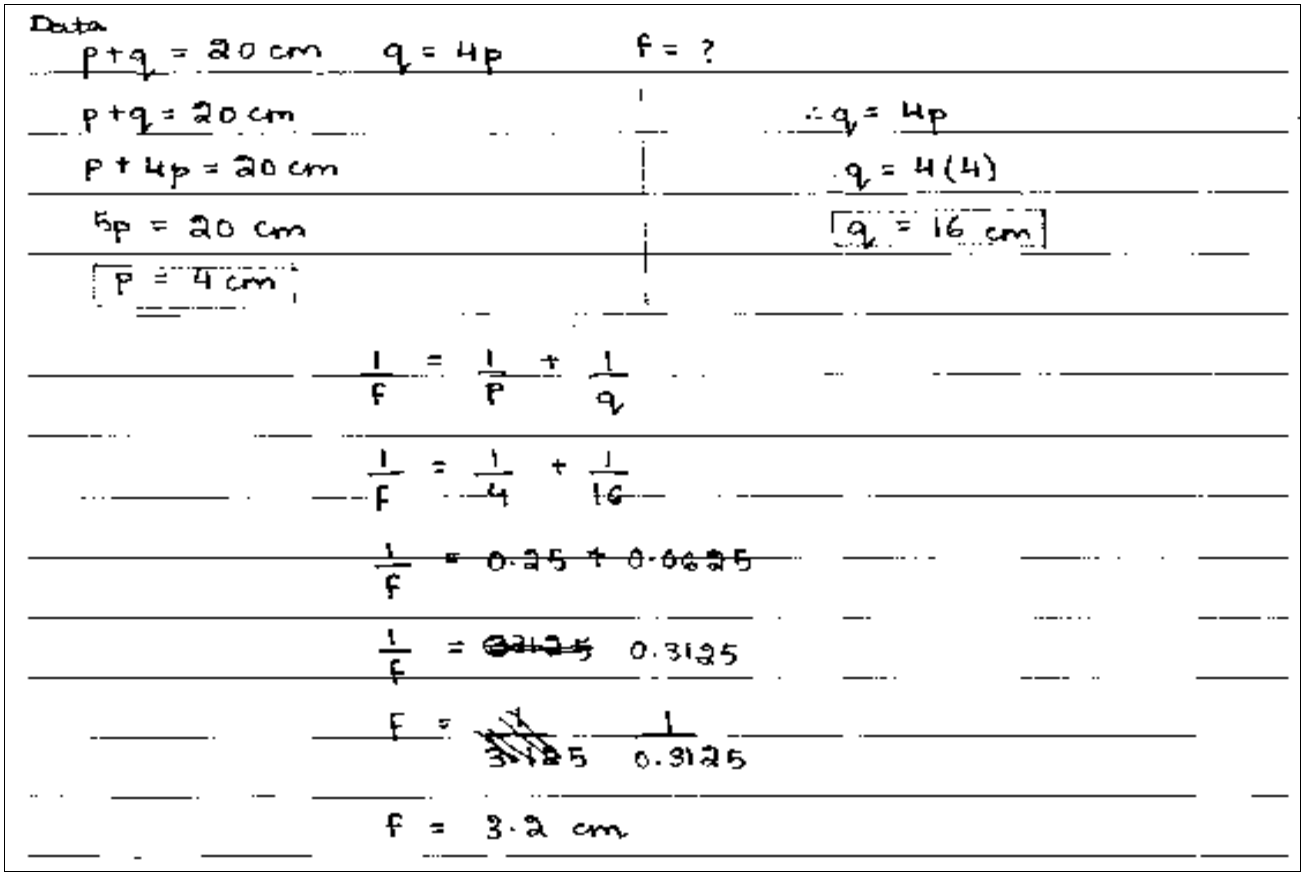
Consider the given ray diagram.



Calculate the value of focal length (f), if (q = 4p).

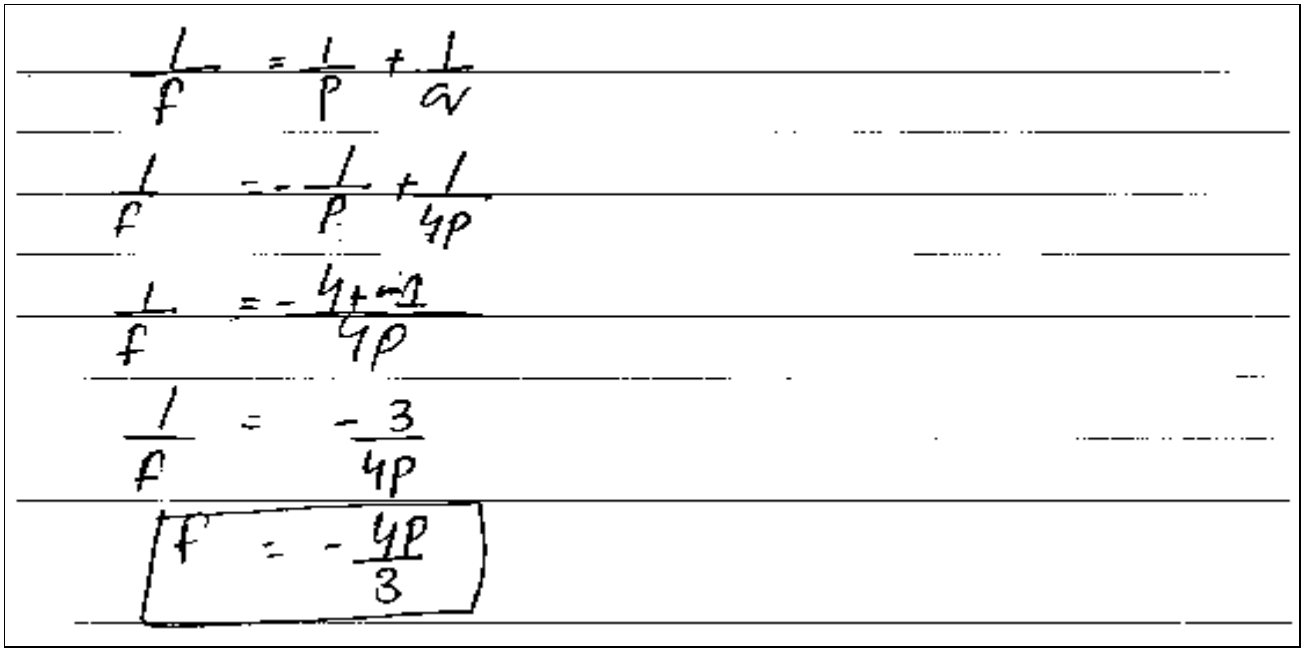
Better responses correctly calculated the value of focal length by extracted data from the given diagram and question. High scoring candidates first found the position of the object (p) and the position of image (q), putting the correct lens formula, 1/f = 1/p + 1/q and calculated the correct value of focal length (f), i.e. 3.2 cm.

Example:



Weaker responses failed to calculate the focal length by the help of the given information in the diagram. They only placed the correct lens formula but were unable to calculate the related quantities like (p), (q) and (f).

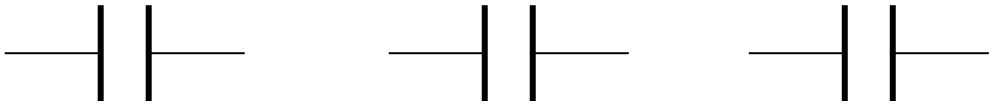
Example:



(**Note**: The question offered a choice between part **a** and part **b**. Majority of the candidates attempted Question **11a** and they scored well in this part.

Question 11a:

Three capacitors of capacitance (C_1), (C_2) and (C_3) are shown in the given figures.



$C_1 = 2\mu F$ $C_2 = 4\mu F$ $C_3 = 8\mu F$

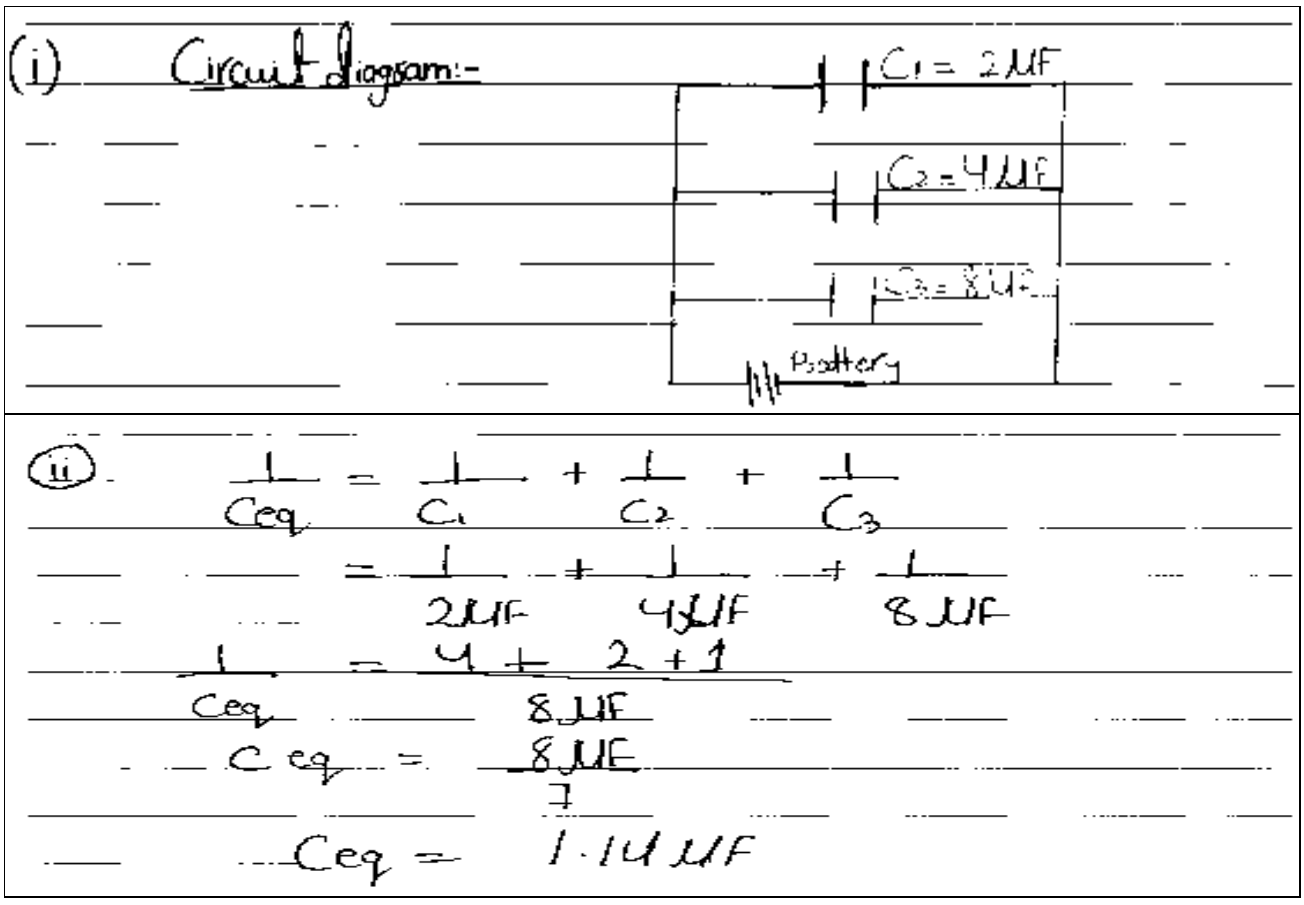
i. Draw a circuit diagram to show the parallel combination of the given capacitors with a power source.

ii. Calculate the effective (equivalent) capacitance of the given capacitors in series combination.

Better responses correctly draw the circuit diagram that shows the parallel combination of the capacitors with a power source as the one shown in the below example of better response image.

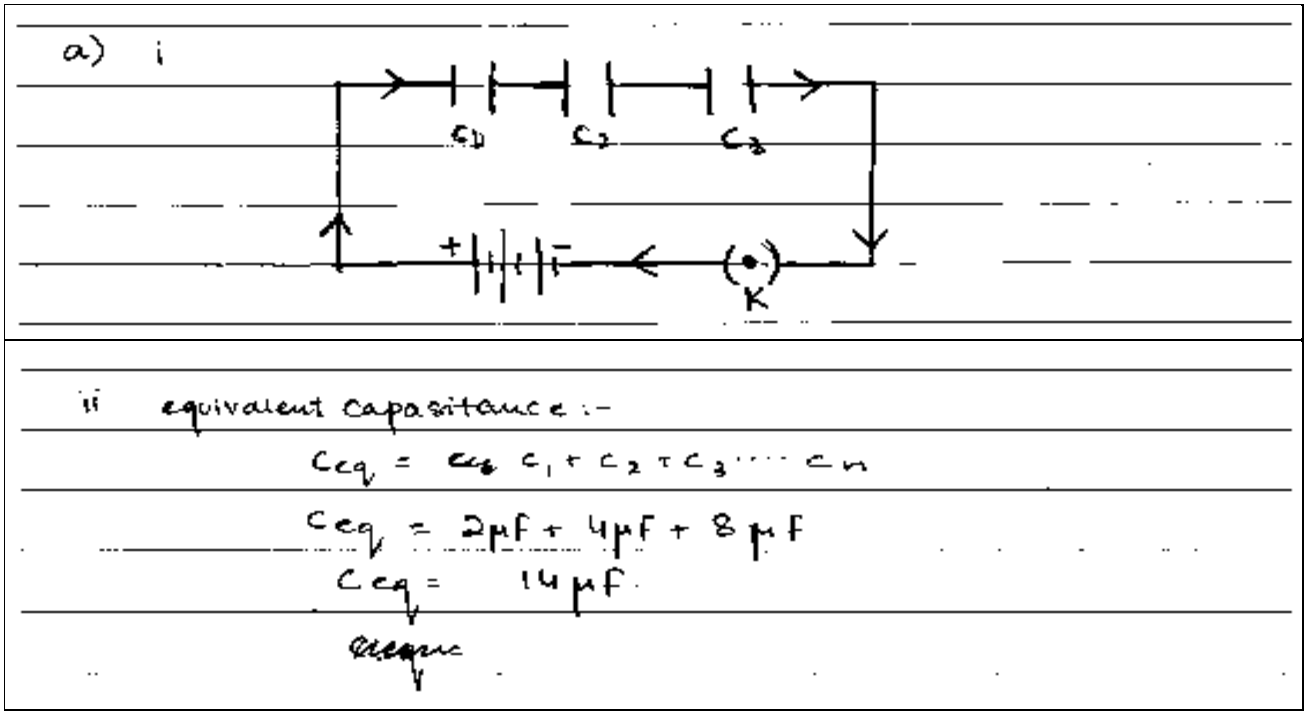
High scoring candidates also calculated the effective capacitance of the capacitors in series combination by putting the correct formula, $\frac{1}{C_e} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}$ and calculated the equivalent capacitance, $C_e = 1.142\ \mu F$.

Example:



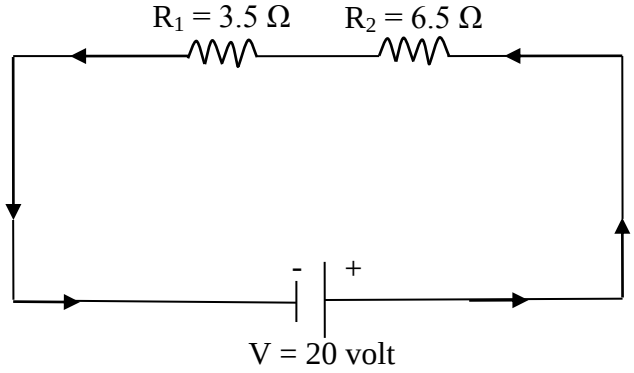
Weaker responses were unable to draw the correct circuit diagram that shows the parallel combination of the capacitors failed to extract the data (values) from the question and incorrectly used the formula of parallel capacitance and therefore, failed to obtain any single mark.

Example:



Question 11b:

The given circuit diagram shows the connection of two different resistors with a battery.



Calculate the following:

i. Effective (equivalent) resistance
ii. Amount of current

Better responses correctly calculated effective (equivalent) resistance and amount of current by extracting data (values) from the question, wrote the correct formula of Ohm's law, i.e. V=I R, added the resistance correctly, i.e. $R_e = R_1 + R_2$ and found the value of current (I) i.e. 2 ampere.

Example:

(b) Data:

Voltage = 20 volts

$R_1 = 3.5\ \Omega$

$R_2 = 6.5\ \Omega$

$I = ?$

$R_{eq} = ?$

Formula:

(i) $R_{eq} = R_1 + R_2$

(ii) $I = \frac{V}{R_{eq}}$

Solution:

(i) $R_{eq} = R_1 + R_2$

$= 3.5 + 6.5$

$R_{eq} = 10\ \Omega$

(ii) $I = \frac{V}{R_{eq}}$

$= \frac{20}{10}$ (∵ series circuit so voltage provided to both resistors is same).

$I = 2$ Amperes. (A).

Ans. Statement:

(i) The equivalent resistance of the circuit is 10 Ω

(ii) The Amount of current flowing through circuit is 2 A (Amperes).

Weaker responses failed to extract the data from the question and were also unable to write the correct formula of Ohm's law. Low scoring candidates wrote the formula of resistance in parallel combination for calculating the value of current (I).

Example:

Effective (equivalent) resistance :-

Formula :-

$$\frac{R_1}{V_1} + \frac{R_2}{V_2} + \frac{R_3}{V_3}$$

$$\frac{3.5}{10} + \frac{6.5}{10} + \frac{R_3}{10}$$

$$\frac{10}{20} + \frac{R_3}{10}$$

$$10(10)20(R_3)$$

$$(100)\ 20R_3$$

$$\frac{100}{120} = R_3$$

$$R_3 = 5$$

$$= \frac{3.5}{10} + \frac{6.5}{10} + \frac{5}{10}$$

$$= \frac{15}{30}$$

Effective (equivalent) resistance is 2.

Amount of current :-

$$= \frac{R_3}{V}$$

$$= \frac{3.5}{10}$$